

Project Journey: Building and Deploying FoodTuck Marketplace

Prepared By: Asif Ali

Introduction

This document outlines the end-to-end process of developing and deploying the FoodTuck marketplace. The project progressed through distinct phases, including planning, design, development, testing, and deployment to a live staging environment on Vercel.

Project Timeline

Day 1: Project Planning & Setup

Tasks Completed:

- Defined project objectives and key features, including product listings, cart functionality, and user profiles.
- Set up the project environment using **Next.js** and integrated **Sanity CMS** for dynamic content management.
- Organized the project folder structure with directories such as `src/`, `public/`, and `components/`.

Tools Used: Visual Studio Code, GitHub, Sanity CMS.

Day 2: Backend Integration

Tasks Completed:

- Integrated APIs to fetch product, category, and user profile data.
- Designed data schemas in **Sanity CMS** for content management.
- Implemented error handling for API calls using `try-catch` blocks.

Challenges Faced:

- Resolved initial API authentication issues by updating tokens.
-

Day 3: Building Core Features

Tasks Completed:

- Developed reusable components for the **product listing, search bar, and filter panel**.
- Implemented **dynamic routing** for individual product pages.
- Managed state using **React Context API**.

Best Practices Followed:

- Ensured **modular component design** and **reusability**.
-

Day 4: Dynamic Frontend Development

Tasks Completed:

- Built responsive UI components ensuring **compatibility across devices**.
- Enhanced UX with **pagination and wishlist functionality**.
- Conducted component testing using mock data.

Performance Optimization:

- Implemented **lazy loading for images**.
 - Minimized unused CSS.
-

Day 5: Testing & Refinement

Tasks Completed:

- Conducted **functional, performance, and security testing**.
- Utilized **Lighthouse** and **GTmetrix** for performance evaluation.
- Fixed UI and API error handling issues.

Results:

- Improved page load time to **under 2 seconds**.
 - Resolved **search bar issues** related to irrelevant query results.
-

Day 6: Deployment Preparation & Staging Setup

Tasks Completed:

- Selected **Vercel** as the hosting platform.
- Connected the GitHub repository to Vercel for **automated builds and deployments**.
- Configured **environment variables** securely within Vercel.
- Deployed the application to a **staging environment** for final testing.

Testing in Staging:

- Verified **product search, cart operations, and checkout workflows**.
- Conducted **cross-browser testing** on Chrome, Firefox, and Edge.

Deployment Process

1. Hosting Platform:

- Utilized **Vercel** for seamless Next.js integration.

2. Repository Setup:

- Organized project files into **logical directories**.

3. Environment Variables:

- Stored sensitive data such as **API keys** securely in Vercel.
- Ensured security by excluding `.env` files from the repository.

4. Deployment Steps:

- Pushed the latest code to the **main branch on GitHub**.
- Vercel automatically triggered a **build** and deployed the site to a live staging URL.

5. Validation:

- Tested for **functionality, responsiveness, and performance**.
- Verified that all features functioned as expected in the **staging environment**.

Conclusion

The FoodTuck marketplace was successfully developed and deployed following industry best practices. The staging environment on Vercel provides a **production-like setting** for ongoing

testing and refinement. This journey highlighted the importance of **structured planning, development best practices, and meticulous testing** in delivering a high-quality application.